

CHEMISTRY**PAPER 1 (THEORY)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 8 printed pages.
4. It is divided into four sections and has twenty one questions in all.
5. Answer all questions.
6. Section A has fourteen subparts. Each question carries 1 mark.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B has ten questions. Each question carries 2 marks.
9. Section C has seven questions. Each question carries 3 marks.
10. Section D has three questions. Each question carries 5 marks.
11. Internal choices have been provided in two questions each in Sections B, C and D.
12. The intended marks for questions are given in brackets [].
13. All working, including rough work, should be done on the same sheet as, and adjacent to the rest of the answer.
14. Balanced equations must be given wherever possible and diagrams where they are helpful.
15. When solving numerical problems, all essential workings must be shown.
16. In working out problems, use the following data:
Gas constant $R = 1.987 \text{ cal deg}^{-1} \text{ mol}^{-1} = 8.314 \text{ J K}^{-1} \text{ mol}^{-1} = 0.0821 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$, $1 \text{ l atm} = 1 \text{ dm}^3 \text{ atm} = 101.3 \text{ J}$.
 $1 \text{ faraday} = 96500 \text{ coulombs}$, Avogadro's number $= 6.023 \times 10^{23}$.

Instruction to Supervising Examiner*Kindly read aloud the Instructions given above to all the candidates present in the examination hall.***SECTION A – 14 MARKS****Question 1**

- (A) Fill in the blanks by choosing the appropriate word(s) from those given in the brackets.
[cinnamic acid, benzoic acid, Perkin, Cannizzaro, increases, decreases, remains constant, Mn^{2+} , MnO_2 , five, three, cumene, toluene, acetone, acetaldehyde, benzoin]
- (i) Benzaldehyde, on heating with acetic anhydride in the presence of sodium acetate, gives _____ [1]
The reaction is known as _____ reaction. (*Recall*)
- (ii) On dilution of an aqueous solution of an electrolyte, its molar conductivity _____ [1]
whereas its conductivity (specific conductance) _____. (*Understand*)

- (iii) In acidic medium, potassium permanganate is reduced to _____ ions, each MnO_4^- ion gaining _____ electrons. (*Understand*) [1]
- (iv) Phenol is manufactured by the air oxidation of _____ (isopropylbenzene) followed by treatment with dilute acid; _____ is obtained as a by-product. (*Recall*) [1]
- (B) Select and write the correct alternative from the choices given below.
- (i) Potassium ferrocyanide, $\text{K}_4[\text{Fe}(\text{CN})_6]$, is 50 % dissociated in a dilute aqueous solution. Its van't Hoff factor is: (*Apply*) [1]
- 1.5
 - 2.5
 - 3.0
 - 2.0
- (ii) Which reaction of glucose shows that its six carbon atoms are linked in a straight chain? (*Analyse*) [1]
- On prolonged heating with HI it gives n-hexane.
 - It reacts with hydroxylamine to form an oxime.
 - It reduces Tollens' reagent.
 - It forms a pentaacetate with acetic anhydride.
- (iii) The coordination number and the oxidation state of cobalt in $[\text{Co}(\text{en})_2(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$ are respectively: (*Apply*) [1]
- 4 and +2
 - 6 and +3
 - 6 and +2
 - 4 and +3
- (iv) Consider the following statements about a zero order reaction $\text{A} \rightarrow \text{products}$: [1]
- P: Its rate is independent of the concentration of A.
 Q: The unit of its rate constant is $\text{mol L}^{-1} \text{s}^{-1}$.
 R: Its half-life is independent of the initial concentration of A.
 S: A plot of $[\text{A}]$ against time is a straight line with slope $-k$.
- Which of the statements are correct?
 (*Evaluate*)
- P and R only
 - Q, R and S only
 - P, Q and R only
 - P, Q and S only
- (v) Reduction of methyl isocyanide (CH_3NC) with lithium aluminium hydride gives: [1]
- (*Understand*)
- ethanamine
 - N-methylmethanamine
 - methanamine
 - ethanamide

- (vi) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

Assertion: The boiling point of ethanol is higher than that of methoxymethane.

Reason: Ethanol has a higher molar mass than methoxymethane.

(Analyse)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Both Assertion and Reason are false.
- (vii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

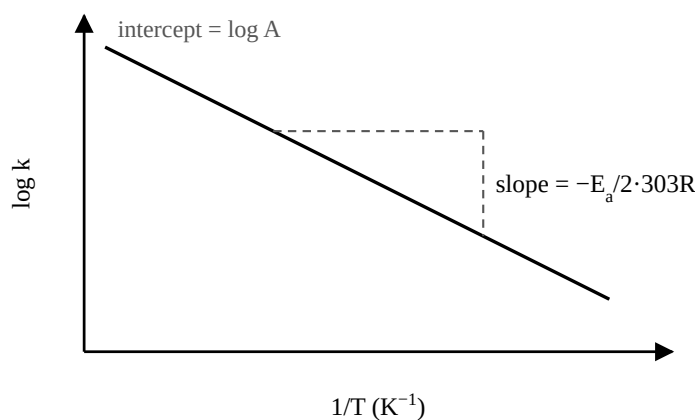
Assertion: Silver liberates hydrogen gas from dilute hydrochloric acid.

Reason: The standard reduction potential of silver is negative.

(Analyse)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Both Assertion and Reason are false.
- (C) Read the passage carefully and answer the questions that follow.

For a reaction to occur, the reacting molecules must collide. According to collision theory, only those collisions in which the molecules have energy equal to or greater than a threshold value and are properly oriented lead to products; these are called effective collisions. The dependence of the rate constant on temperature is given by the Arrhenius equation, $k = Ae^{-E_a/RT}$, or $\log k = \log A - E_a/2 \cdot 303RT$. A plot of $\log k$ against $1/T$ is therefore a straight line. A catalyst provides an alternative path of lower activation energy; it increases the rate of the forward and the backward reaction equally.



Arrhenius plot

- (i) For a reaction, the slope of the plot of $\log k$ against $1/T$ is -5000 K . Calculate the activation energy of the reaction. (Apply) [1]

- (ii) Why do all collisions between reactant molecules not result in the formation of products? [1]
(Understand)
- (iii) How does a catalyst increase the rate of a reaction? Does it change the enthalpy of the reaction? (Analyse) [1]

SECTION B – 20 MARKS

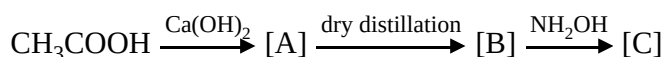
Question 2

The standard emf of the Daniell cell, $\text{Zn} | \text{Zn}^{2+}(1 \text{ M}) || \text{Cu}^{2+}(1 \text{ M}) | \text{Cu}$, is 1.10 V. State what happens (direction of electron flow and the reactions at the electrodes) when an external opposing potential is applied of:

- (i) 1.0 V (Analyse) [1]
- (ii) 1.2 V (Analyse) [1]

Question 3

Identify the compounds [A], [B] and [C] in the following sequence and write the equation for the formation of [C] from [B]: [2]



(Understand)

Question 4

- (i) Answer the following questions for $[\text{Ni}(\text{CO})_4]$ (atomic number of Ni = 28). [1]
- (a) What is the oxidation state and the hybridisation of nickel? (Understand) [1]
- (b) State its geometry and magnetic nature. (Analyse) [1]

OR

- (ii) Using crystal field theory, answer the following for $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{CN})_6]^{4-}$ (atomic number of Fe = 26). [1]
- (a) Write the electronic configuration of the metal ion in each in terms of t_{2g} and e_g . (Understand) [1]
- (b) How many unpaired electrons does each have? (Analyse) [1]

Question 5

A first order reaction takes 40 minutes for 30 % decomposition. Calculate its half-life. (Apply) [2]

Question 6

How can the following conversions be carried out? (Write the chemical equations.)

- (i) Aniline to acetanilide (Recall) [1]
- (ii) Ethanal to propan-2-ol (Recall) [1]

Question 7

Substantiate the following statements with reasons.

- (i) Transition elements show variable oxidation states; manganese (Z = 25) shows all states from +2 to +7. (Understand) [1]
- (ii) Cr^{2+} is a reducing agent whereas Mn^{3+} is an oxidising agent, though both are d^4 . (Atomic number of Cr = 24, Mn = 25) (Analyse) [1]

Question 8

An aqueous solution of urea boils at $100.128\text{ }^{\circ}\text{C}$. Calculate its freezing point. (K_b of water = $0.512\text{ K kg mol}^{-1}$, K_f of water = $1.86\text{ K kg mol}^{-1}$) (*Apply*) [2]

Question 9

(i) Illustrate the following reactions with an example in each case:

(a) Diazotisation (*Recall*) [1]

(b) Balz–Schiemann reaction (*Recall*) [1]

OR

(ii) Propanoic acid on heating with ammonia gives [A], $\text{C}_3\text{H}_7\text{NO}$. [A] on warming with bromine and KOH gives [B]. [B] reacts with nitrous acid to give [C] with the evolution of nitrogen, and gives a foul-smelling compound [D] when warmed with chloroform and alcoholic KOH. Write the structures and names of [A], [B], [C] and [D]. (*Understand*) [2]

Question 10

The half-life of a first order reaction is 40 minutes at 300 K and 20 minutes at 320 K. Calculate the energy of activation of the reaction. (*Apply*) [2]

Question 11

A sample of 250 g of sea water from a coastal lagoon contained 1.17 g of sodium chloride and boiled at 373.0806 K (assume NaCl is the only solute). Calculate the observed molar mass of NaCl in the sample and its van't Hoff factor. (Given: K_b of water = $0.52\text{ K kg mol}^{-1}$, boiling point of water = 373 K , molar mass of NaCl = 58.5 g mol^{-1}) (*Analyse*) [2]

SECTION C – 21 MARKS**Question 12**

(i) 2-Bromobutane is optically active. Draw its two enantiomers and state the one property in which they differ. (*Understand*) [1]

(ii) An optically active sample of 3-bromo-3-methylhexane is hydrolysed by aqueous KOH by the $\text{S}_{\text{N}}1$ mechanism. Riya says that the product is optically active; Kabir says that it is a racemic mixture. Who is correct? Explain. (*Evaluate*) [1]

(iii) Write the equation for the reaction of 2-bromobutane with alcoholic KOH and name the rule that decides the major product. (*Apply*) [1]

Question 13

A solution of 0.5 g of KCl in 100 g of water freezes at $-0.24\text{ }^{\circ}\text{C}$. (K_f of water = $1.86\text{ K kg mol}^{-1}$, molar mass of KCl = 74.5 g mol^{-1})

(i) Calculate the observed molar mass of KCl. (*Apply*) [1]

(ii) Calculate the van't Hoff factor. (*Apply*) [1]

(iii) Calculate the degree of dissociation of KCl. (*Analyse*) [1]

Question 14

(i) An alcohol [A], $\text{C}_4\text{H}_{10}\text{O}$, gives turbidity immediately with Lucas reagent at room temperature.

(a) Identify [A] and write the equation for the reaction. (*Analyse*) [1]

- (b) Why do tertiary alcohols react fastest with Lucas reagent? (*Understand*) [1]
- (ii) Phenol undergoes electrophilic substitution much more readily than benzene. Explain. (*Understand*) [1]

OR

- (iii) How will the following be converted? (Give chemical equations.)
- (a) Phenol to picric acid (*Recall*) [1]
- (b) Phenol to phenyl acetate (*Recall*) [1]
- (c) Ethanol to ethanal (*Recall*) [1]

Question 15

- (i) Complete and balance: $\text{KMnO}_4 + \text{FeSO}_4 + \text{H}_2\text{SO}_4 \rightarrow \text{---} + \text{---} + \text{---} + \text{---}$ (*Understand*) [1]
- (ii) Complete and balance: $\text{K}_2\text{Cr}_2\text{O}_7 + \text{Na}_2\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{---} + \text{---} + \text{---} + \text{---}$ (*Understand*) [1]
- (iii) Why do actinoids show a larger number of oxidation states than lanthanoids? (*Analyse*) [1]

Question 16

- (i) Name the products of hydrolysis of maltose. Is maltose a reducing sugar? (*Recall*) [1]
- (ii) What is a zwitterion? Write the zwitterionic form of glycine. (*Understand*) [1]
- (iii) State two differences between DNA and RNA. (*Understand*) [1]

Question 17

- (i) Write the chemical equations for the following:
- (a) Chloroform is heated with silver powder. (*Recall*) [1]
- (b) Chlorobenzene to aniline (*Recall*) [1]
- (c) Wurtz–Fittig reaction (*Recall*) [1]

OR

- (ii) Answer the following:
- (a) Arrange in decreasing order of reactivity towards $\text{S}_{\text{N}}2$ substitution: 2-bromo-2-methylpropane, 1-bromobutane, bromomethane, 2-bromobutane. (*Analyse*) [1]
- (b) Why are haloarenes much less reactive than haloalkanes towards nucleophilic substitution? (*Understand*) [1]
- (c) Chlorine deactivates the benzene ring, yet chlorobenzene gives *ortho* and *para* products on nitration. Explain. (*Evaluate*) [1]

Question 18

Write the chemical equations for the following conversions: (*Recall*)

- (i) Benzene to aniline [1]
- (ii) Aniline to 4-bromoaniline [1]
- (iii) Ethanenitrile to ethanoic acid [1]

SECTION D – 15 MARKS

Question 19

- (i) An organic compound [A], $C_3H_6O_2$, liberates CO_2 with sodium hydrogencarbonate. [A] reacts with chlorine in the presence of red phosphorus to give [B], $C_3H_5ClO_2$. [A] with thionyl chloride gives [C], which on hydrogenation over palladium on barium sulphate gives [D], C_3H_6O . [D] reduces Tollens' reagent but does not give the iodoform test.
- (a) Identify [A], [B], [C] and [D]. (*Understand*) [2]
- (b) Write the equation for the conversion of [C] into [D] and name the reaction. (*Recall*) [1]
- (c) Which is the stronger acid, [A] or [B]? Why? (*Evaluate*) [1]
- (d) Write the structure and IUPAC name of the product formed when [D] is treated with dilute NaOH. (*Apply*) [1]

OR

- (ii) Answer the following:
- (a) Give a chemical test to distinguish between: (1) methanal and ethanal (2) phenol and benzoic acid. (*Analyse*) [2]
- (b) Methanal does not undergo aldol condensation but undergoes the Cannizzaro reaction. Why? (*Understand*) [1]
- (c) Write chemical equations to convert: (1) ethanal to 2-hydroxypropanoic acid (2) benzaldehyde to benzoin. (*Recall*) [2]

Question 20

- (i) For the complex $[Co(NH_3)_4Cl_2]Cl$:
- (a) Write its IUPAC name. (*Recall*) [1]
- (b) Draw the geometrical isomers of the complex ion. Is either of them optically active? (*Analyse*) [1]
- (ii) Name the type of isomerism shown by each pair: (a) $[Co(NH_3)_5(NO_2)]Cl_2$ and $[Co(NH_3)_5(ONO)]Cl_2$ (b) $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$ (*Analyse*) [1]
- (iii) A complex of composition $PtCl_4 \cdot 4NH_3$ gives two moles of AgCl per mole with silver nitrate, and its molar conductance corresponds to three ions. Write its structural formula and IUPAC name. (*Evaluate*) [1]
- (iv) $[Ti(H_2O)_6]^{3+}$ is violet. Explain its colour, and state how the wavelength absorbed would change if water were replaced by a stronger-field ligand. (*Analyse*) [1]

Question 21

- (i) Answer the following:
- (a) Calculate the emf of the following cell at 298 K: [2]
- $$Mg(s) | Mg^{2+}(0.130 M) || Ag^+(1.0 \times 10^{-4} M) | Ag(s)$$
- Given: $E^\circ(Mg^{2+}/Mg) = -2.37 V$, $E^\circ(Ag^+/Ag) = +0.80 V$ (*Apply*)
- (b) Calculate ΔG for the cell reaction in (a). (*Apply*) [1]
- (c) How many coulombs are required to reduce 1 mol of MnO_4^- to Mn^{2+} ? (*Apply*) [1]

(d) Write the anode and cathode reactions in the rusting of iron. (*Understand*) [1]

OR

(ii) Answer the following:

(a) A current of 5.0 A is passed for 1 hour 20 minutes through molten Al_2O_3 (dissolved in cryolite). Calculate the mass of aluminium deposited at the cathode. (Atomic mass of Al = 27) [2]
(*Apply*)

(b) Draw the plots of molar conductivity against $\sqrt{c}$ for KCl and for acetic acid. Why can the limiting molar conductivity of acetic acid not be found from its plot? (*Analyse*) [2]

(c) Write the electrode reactions of the hydrogen–oxygen fuel cell and state one advantage of it. [1]
(*Recall*)

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